

# Operator product expansion algebra

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based on joint work with M. Fröb, J. Holland and Ch. Kopper

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# Technical challenges of QFT

Unfortunately, if they mathematically exist, QFTs must be rather complicated presumably in any approach/framework.

*“In those years a theoretical physicist of modest talent could harvest results of great interest, today persons of great talent produce results of modest interest”* [Hund]

## Reasons

- ▶ Basic source of trouble:  $\mathcal{O}(x)$  at a sharp point  $x$  is a singular object since it creates states with arbitrary high energy from vacuum.
- ▶ Relativistic invariance + Hilbert-space positivity  $\Rightarrow$  correlation fcts.  $\langle 0 | \mathcal{O}(x_1) \mathcal{O}(x_2) \dots | 0 \rangle$  always singular [Lehmann 1954]
- ▶ Impossible to identify in  $\mathcal{H}$  subspaces associated with a definite localization in  $x$ -space: Vectors  $\int d^4x f(x) \mathcal{O}(x) | 0 \rangle$  span entire Hilbert space! [Reeh-Schlieder 1968]
- ▶  $\mathcal{O}(x) | 0 \rangle$  contains arbitrarily many particles when there is interaction  $\Rightarrow$  worse than in non-relativistic  $N$ -body systems

## Mathematical status of QFT

The inherent technical complications have so far strongly impeded progress establishing mathematical existence of interesting QFTs in  $d = 4$  dimensions. One of the sources of trouble is the need for renormalization.

Initially “**problem**”:

*I must say that I am very dissatisfied with the situation, because this so-called ‘good theory’ does involve neglecting infinities which appear in its equations, neglecting them in an arbitrary way [...] Sensible mathematics involves neglecting a quantity when it is small not neglecting it just because it is infinitely great and you do not want it!* [Dirac 1975]

Later “**feature**”:

*The renormalization method is [...] connecting theories describing physical processes at different length scales and thereby connecting different theories in the physical sciences* [Kadanoff 2000]

# History of renormalization

- ▶ 30s: Dirac, Born-Heisenberg-Jordan, Uehling and others discover that many calculations in QFT lead to infinite results.
- ▶ 40s: Bethe, Kramers, Weisskopf, Feynman, Schwinger, Tomonaga, and others show how to “hide” infinities in physical parameters
- ▶ 50s: Dyson, Petermann-Stückelberg, Gell-Mann-Low, Bogoliubov-Shirkov: General understanding of “multiplicative” nature of renormalization, power counting
- ▶ 60s: Hepp’s analysis of Feynman integrals, Zimmermann forest formula
- ▶ 70s: Callan-Symanzik renormalization group equations, Wegner-Houghton functional equation, Kadanoff-Wilson picture of renormalization group flow, Epstein-Glaser method, Gross-Wilcek, Politzer: asymptotic freedom
- ▶ 80s: Polchinski functional equation,
- ▶ 90s: Wetterich functional equation, numerical studies
- ▶ 00s: Brunetti-Fredenhagen, Hollands-Wald: Renormalization on curved Lorentzian spacetimes

# Operator product expansion: History

- ▶ 1969 Wilson proposes **O**perator **P**roduct **E**xpansion (OPE)
- ▶ 1970 Zimmermann shows OPE consistent with renormalized perturbation theory [Brandeis lectures 1970]
- ▶ 1972 Wilson-Zimmermann: general arguments in favor of OPE
- ▶ In 1970s, various groups [Polyakov 1974, Mack 1977, Gatto et al. 1973, Schroer et al. 1974] realize that the OPE simplifies in CFTs and associativity constraints can be turned into “conformal bootstrap”.
- ▶ In 1980s, OPE to study conformal field theories in  $d = 2$  [Belavin et al. 1984].
- ▶ Borchers and others propose to formalize their ideas in the framework of Vertex Operator Algebras [Borchers 1988]  $\Rightarrow$  Mathematics
- ▶ Recently numerical bootstrap: new approach to CFTs in  $d > 2$  [Rychkov 2016],...
- ▶ This talk: Variational principle and OPE, convergence questions (mostly perturbation theory)

# Formulating QFT via operator product expansion

An intrinsically “generally covariant” formulation of QFT can be given via algebraic methods, e.g. by formulating QFT via OPE [Hollands-Wald 2012]. A quantum field theory consists of:

- ▶ A list of quantum fields  $\{\mathcal{O}_A\}$ , where  $A$  is a label (incl. tensor/spinor indices)
- ▶ A state  $\Psi$  is an expectation value functional characterized by  $N$ -point “functions”  $\langle \mathcal{O}_{A_1}(x_1) \dots \mathcal{O}_{A_N}(x_N) \rangle_\Psi$ . Such a functional should be “positive”  $\rightarrow$  probability interpretation!
- ▶  $N$ -point functions should satisfy a “micro local spectrum condition”
- ▶ The OPE should hold for a wide class of states  $\Psi$

$$\langle \mathcal{O}_{A_1}(x_1) \dots \mathcal{O}_{A_N}(x_N) \rangle_\Psi = \sum_B \underbrace{c_{A_1 \dots A_N}^B(x_1, \dots, x_N)}_{\text{OPE coefficients}} \langle \mathcal{O}_B(x_N) \rangle_\Psi$$

- ▶ The OPE coefficients are independent of  $\Psi$ .
- ▶ The OPE coefficients should be generally covariant functionals of the metric  $g_{\mu\nu}$ .
- ▶ The OPE should satisfy associativity law.

## Example: Free field

For a free scalar field theory in  $d = 4$  dimensions with action  $\int |\partial\phi|^2$ , the basic OPE relation is

$$\phi(x_1)\phi(x_2) = \frac{\lambda}{|x_1 - x_2|^2} \cdot 1 + \underbrace{\phi^2(x_2) + \sum \frac{(x_1 - x_2)^{\mu_1} \dots (x_1 - x_2)^{\mu_N}}{N!} \phi \partial_{\mu_1 \dots \mu_N} \phi(x_2)}_{\text{smooth part}} \quad (0.1)$$

The composite fields such as  $\mathcal{O} = \phi^2$  are defined by this equation. Other composite fields  $\mathcal{O} = \phi^4, \phi^3 \nabla_\mu \phi, \dots$  similarly occur in OPE of  $\phi^2$ , etc. Everything is constrained by associativity.

So in this theory one has, e.g.

$$C_{AB}^C = \frac{\lambda}{|x_1 - x_2|^2}$$

when  $\mathcal{O}_A = \mathcal{O}_B = \phi, \mathcal{O}_C = 1$ , etc. In curved spacetime the distances  $|x_1 - x_2|$  in the coefficients are replaced by geometric quantities related to the theory of geodesics.

## Example: Free field

Curved spacetime  $g_{\mu\nu}$ :

$$\begin{aligned} \phi(x_1)\phi(x_2) = & \frac{\lambda_1 + \lambda_2 R_{\mu\nu} \nabla^\mu \sigma_{12} \nabla^\nu \sigma_{12} + \dots}{\sigma_{12}} \cdot 1 \\ & + \underbrace{\phi^2(x_2) + \sum \frac{\nabla^{\mu_1} \sigma_{12} \dots \nabla^{\mu_N} \sigma_{12}}{N!} \phi \nabla_{\mu_1 \dots \mu_N} \phi(x_2)}_{\text{smooth part}} \end{aligned} \quad (0.2)$$

$\sigma$  = squared geodesic distance. So in this theory one has, e.g.

$$\mathcal{C}_{AB}^C = \frac{\lambda_1 + \lambda_2 R_{\mu\nu} \nabla^\mu \sigma_{12} \nabla^\nu \sigma_{12} + \lambda_3 R \sigma_{12} \log \sigma_{12} + \dots}{\sigma_{12}}$$

when  $\mathcal{O}_A = \mathcal{O}_B = \phi$ ,  $\mathcal{O}_C = 1$ , etc. So OPE coefficients are built from theory-dependent constants  $\lambda_1, \lambda_2, \dots$  quantities built just from metric  $R, R_{\mu\nu}, \sigma_{12}, \dots$

## Example: Conformal field theory (CFT)

In conformal field theory ( $d = 4$ ) on flat spacetime  $\mathbb{R}^4$ , it is natural to group composite fields into “multiplets” transforming under the conformal group  $O(4, 2)$ . Each multiplet contains a “primary field”  $\mathcal{O}$ , together with its “descendants”, which are roughly given by  $\partial_{\mu_1} \dots \partial_{\mu_N} \mathcal{O}$ .

E.g.  $\phi^2$  is a primary field,  $\phi \partial_\mu \phi$  a descendant. The OPE between two primary fields  $\mathcal{O}_A, \mathcal{O}_B$  takes the form

$$\mathcal{O}_A(x_1) \mathcal{O}_B(x_2) = \sum_{\text{primary } C} \frac{\lambda_{AB}^C}{|x_1 - x_2|^{\Delta_A + \Delta_B - \Delta_C}} \mathcal{P}(x_1 - x_2, \partial) \mathcal{O}_C(x_2)$$

where  $\mathcal{P} = \mathcal{P}_{AB}^C$  is a (pseudo-) differential operator that is determined completely by group theoretical considerations [Schroer & Swieca 1974]. Thus the content of the theory is determined by (i) structure constants  $\lambda_{AB}^C$  and (ii) dimensions  $\Delta_A$ .

Associativity+OS positivity put very stringent conditions on these data  $\rightarrow$  conformal bootstrap [Mack 1977, Polyakov 1974, Dolan-Osborne 2000,..., present].

It is natural to ask:

## This talk

1. How to compute OPE coefficients  $\mathcal{C}_{AB...}^C$  (even in principle) beyond free field or CFTs?
2. In what sense does associativity hold in general?
3. What is the magnitude of the “remainder” in the OPE (=error term)?
4. Can one devise an axiomatic framework for QFT in terms of OPE?

In this talk, I will give some answers to these questions.

# Outline

**1** How to construct the OPE coefficients

**2** OPE factorisation

**3** OPE convergence

# General idea

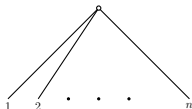
- ▶ For free field theories (e.g. free scalar field theory with action  $S_{CFT} = \int |\partial\phi|^2$ ) one can construct OPE directly by “Wick’s theorem.”
- ▶ CFTs in  $d = 2$ , one can use representation theoretic methods (Virasoro-algebra,  $W$ -algebras, current algebras, ...)
- ▶ For CFTs in  $d > 2$  dimensions, one can use conformal bootstrap including its numerical versions [Polyakov, Mack, Gatto et al., ..., Rychkov et al., ...]
- ▶ Some progress has been made for lattice QFTs (numerical) [Monahan et al. 2013, 2014]
- ▶ For perturbations of free field theories or CFTs (given intuitively by  $S_{CFT} + g \int \mathcal{O}$ , where  $\mathcal{O}$  is some “marginal” or “relevant” operator), one can attempt to derive a differential equation for the OPE coefficients  $\mathcal{C}_{AB...C}^D$  as a function of the coupling  $g$ .
- ▶ This type of equation was found (and proved) by [Holland & Holland 2014], generalizing and correcting an earlier attempt by [Guida & Magnoli 1995].

# Action principle

To write down the action principle, use **graphical notation**. I draw an OPE coefficient

$$\mathcal{C}_{A_1 \dots A_n}^B(x_1, \dots, x_n)$$

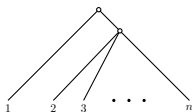
as



I draw a concatenation of OPE coefficients

$$\mathcal{C}_{A_1 C}^B(x_1, x_n) \mathcal{C}_{A_2 \dots A_n}^C(x_2, \dots, x_n)$$

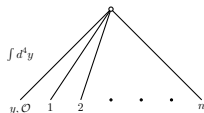
as



Attention: None of these diagrams is a “Feynman graph”!

# Action principle

I also write

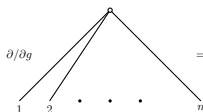


where

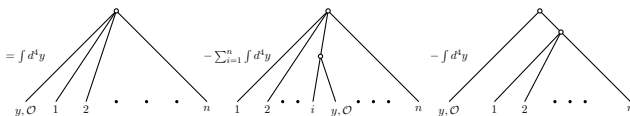
- ▶  $\mathcal{O}$  denotes the “deformation”
- ▶  $\int d^4 y = \text{integral over } \{|y - x_n| < L\}$ .
- ▶  $L = \text{length scale that is part of the definition of the theory.}$

# Action principle

There is a kind of “action principle” for OPE coefficients if we “deform”  
 $S_{CFT} \rightarrow S_{CFT} + g \int \mathcal{O}$ :



**Figure:** Functional equation, left side. The tree represents a coefficient  
 $\mathcal{C}_{A_1 \dots A_n}^B(x_1, \dots, x_n)$



**Figure:** Functional equation, right side. The composite trees represent concatenations of coefficients, e.g. the rightmost tree means  
 $\sum_C \mathcal{C}_{A_1 \dots A_n}^C(x_1, \dots, x_n) \mathcal{C}_{OC}^B(y, x_n)$

# Action principle

## Theorem (Hollands-JH)

To any order in  $g$ :

$$\begin{aligned}\partial_g \mathcal{C}_{A_1 \dots A_N}^B(x_1, \dots, x_N) = & - \int_{|y-x_N| < L} d^4 y \left[ \mathcal{C}_{\mathcal{O}_{A_1 \dots A_N}}^B(y, x_1, \dots, x_N) \right. \\ & - \sum_{i=1}^N \sum_{[C] \leq [A_i]} \mathcal{C}_{\mathcal{O}_{A_i}}^C(y, x_i) \mathcal{C}_{A_1 \dots \widehat{A_i} C \dots A_N}^B(x_1, \dots, x_N) \\ & \left. - \sum_{[C] < [B]} \mathcal{C}_{A_1 \dots A_N}^C(x_1, \dots, x_N) \mathcal{C}_{\mathcal{O}_C}^B(y, x_N) \right].\end{aligned}$$

- ▶ Can compute OPE coefficients to any perturbation order by iteration.
- ▶ State independence obvious.
- ▶  $L \rightarrow L'$  equivalent to

$$\mathcal{O}_A \rightarrow \mathcal{O}'_A = \sum Z_A^B(g, \tau) \cdot \mathcal{O}_B \quad (1.3)$$

and  $g \rightarrow g' = g(g, \tau) \Rightarrow$  RG equations! ( $\tau = \log L/L' =$  RG “time”).

# Examples

- ▶ In  $\phi^4$  theory ( $d = 4$ ), i.e.  $\mathcal{O} = -\phi^4$ , one can compute OPE coefficients order by order. At each order, one  $d^4y$ -integral  $\Rightarrow$  at order  $g^r$  we have  $r$  integrations  $\Rightarrow$  similar complexity as “Feynman diagram” method. But: Renormalization “automatic”.
- ▶ For Gross-Neveu model ( $d = 2$ ), i.e.  $\mathcal{O} = -(\bar{\psi}\psi)^2$ , sharp (?) all order bounds on OPE coefficients. [Hollands & Holland, in prep.]  $\Rightarrow$  can analyze convergence of series in  $g$ .
- ▶ For marginal perturbations of CFTs, simplification of equation to ODE.
- ▶ For local gauge theories (e.g. YM-theory), there holds a similar action principle, supplemented by an “evolution equation” for the BRST-operator (as a function of  $g$ ) [Fröb 2016]

# Action principle in CFT

If I assume to be given a 1-parameter families of CFTs with an exactly marginal operator  $\mathcal{O}$  (i.e.  $\Delta_{\mathcal{O}} = d$  in  $d$  dimensions) parameterized by  $g$ , then action principle implies an equation of the form

$$\begin{aligned}\frac{d}{dg}\lambda &= f_{\mathcal{O}}^{\lambda}(\Delta, \lambda) \\ \frac{d}{dg}\Delta &= f_{\mathcal{O}}^{\Delta}(\Delta, \lambda)\end{aligned}\tag{1.4}$$

where  $f_{\mathcal{O}}^{\Delta}, f_{\mathcal{O}}^{\lambda}$  are explicit (quadratic) functions that depend on  $6j$ -symbols of the group  $O(4, 2)$  in  $d = 4$  (i.e. entirely group theoretic=kinematic). Here  $\lambda = \{\lambda_{AB}^C(g)\}$  and  $\Delta = \{\Delta_A(g)\}$  are the CFT data which are now functions of  $g$ .  $\mathcal{O}$  is the (marginal) perturbation of the CFT, which enters the functions. [Hollands, in prep.]

# Outline

1 How to construct the OPE coefficients

2 OPE factorisation

3 OPE convergence

# The OPE factorises

## Theorem (Holland-SH)

*In  $\phi^4$ -theory, any arbitrary but fixed loop order:*

$$\mathcal{C}_{A_1 \dots A_N}^B(x_1, \dots, x_N) = \sum_C \mathcal{C}_{A_1 \dots A_M}^C(x_1, \dots, x_M) \mathcal{C}_{C A_{M+1} \dots A_N}^B(x_M, \dots, x_N)$$

*holds on the domain  $\xi \equiv \frac{\max_{1 \leq i \leq M} |x_i - x_M|}{\min_{M < j \leq N} |x_j - x_M|} < 1$ . (Sum over  $C$  abs. convergent !)*

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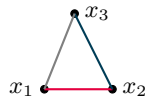
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For  $N = 3$ :  $\xi = \frac{|x_1 - x_2|}{|x_2 - x_3|} < 1$



for  $\xi \ll 1$



for  $\xi \approx 1$

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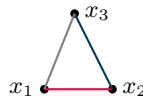
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This shows associativity really holds!

- Bound on remainder
- Justification of “action principle”

# Quantitative bound

## Theorem

*Up to any perturbation order  $r \in \mathbb{N}$  the bound*

$$\begin{aligned} & \left| \text{Remainder in associativity} \right| \\ & \leq \frac{K_r \xi^{D+1} \max_{N \leq v < N} |x_i - x_n|^{[B]}}{\prod_{v=1}^M \min_{1 \leq w \leq M, w \neq v} |x_v - x_w|^{[A_v] + \delta} \prod_{i=M+1}^N \min_{M < j \leq N, i \neq j} |x_i - x_j|^{[A_i] + \delta}} \end{aligned}$$

*holds for some  $\delta > 0$  and where*

$$\xi := \frac{\max_{1 \leq i \leq M} |x_i - x_M|}{\min_{M < j \leq N} |x_j - x_M|}$$

*and where  $K_r$  is a constant which does not depend on  $D$ . (Here  $[A] = \text{dim. of op. in free theory}$ ).*

# Outline

1 How to construct the OPE coefficients

2 OPE factorisation

**3 OPE convergence**

## Previous results

- ▶ Wilson proposed his expansion as an asymptotic expansion for short distances
- ▶ In CFTs, Mack showed convergence (in  $d = 4$ ) for finite distances [Mack 1977]; for a more formal argument see also [Pappadopulo et al. 2012]
- ▶ There is a difference between space like separation and light like separation
- ▶ For theories without conformal invariance, situation was unclear

# Bound on OPE remainder I

## Theorem (Holland-Kopper-SH)

At any perturbation order  $r$  and for any  $D \in \mathbb{N}$ ,

$$\left| \overbrace{\left\langle \left( \mathcal{O}_{A_1}(x_1) \cdots \mathcal{O}_{A_N}(x_N) - \sum_{\dim[B] \leq D} \mathcal{C}_{A_1 \dots A_N}^B(x_1, \dots, x_N) \mathcal{O}_B(x_N) \right) \underbrace{\hat{\varphi}(p_1) \cdots \hat{\varphi}(p_n)}_{\text{Spectator fields}} \right\rangle}_{\text{OPE-Remainder}} \right|$$

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## Theorem (Holland-Kopper-SH)

At any perturbation order  $r$  and for any  $D \in \mathbb{N}$ , there exists a  $K > 0$  such that

$$\begin{aligned} & \overbrace{\left| \left\langle \left( \mathcal{O}_{A_1}(x_1) \cdots \mathcal{O}_{A_N}(x_N) - \sum_{\dim[B] \leq D} \mathcal{C}_{A_1 \dots A_N}^B(x_1, \dots, x_N) \mathcal{O}_B(x_N) \right) \underbrace{\hat{\varphi}(p_1) \cdots \hat{\varphi}(p_n)}_{\text{Spectator fields}} \right\rangle \right|}_{\text{OPE-Remainder}} \\ & \leq \frac{M^{n-1}}{\sqrt{D!}} \frac{\left( KM \max_{1 \leq i \leq N} |x_i - x_N| \right)^{D+1}}{\min_{1 \leq i < j \leq N} |x_i - x_j|^{\sum_i \dim[A_i] + 1}} \cdot \sup \left( 1, \frac{|P|}{\sup(m, \kappa)} \right)^{(D+2)(r+5)} \end{aligned}$$

►  $M = \begin{cases} m & \text{for } m > 0 \\ \mu & \text{for } m = 0 \end{cases}$  mass or renormalization scale

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- ▶  $|P| = \sup_i |p_i|$ : maximal momentum of spectators

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- ▶  $|P| = \sup_i |p_i|$ : maximal momentum of spectators
- ▶  $\kappa := \inf(\mu, \varepsilon)$ , where  $\varepsilon = \min_{I \subset \{1, \dots, n\}} |\sum_I p_i|$   
 $\varepsilon$ : distance of  $(p_1, \dots, p_n)$  to “exceptional” configurations

## Conclusions from bound on OPE remainder

$$\text{"OPE remainder"} \leq \frac{M^{n-1}}{\sqrt{D!}} \frac{\left(K M \max_{1 \leq i \leq N} |x_i - x_N|\right)^{D+1}}{\min_{1 \leq i < j \leq N} |x_i - x_j|^{\sum_i \dim[A_i] + 1}} \cdot \sup \left(1, \frac{|P|}{\sup(m, \kappa)}\right)^{(D+2)(r+5)}$$

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1. Massive fields ( $m > 0$ ): Bound is finite for arbitrary  $p_1, \dots, p_n$
2. Massless fields: Bound is finite only for non-exceptional  $p_1, \dots, p_n$   
(compare [Lowenstein-Zimmermann 1976])

# Conclusions from bound on OPE remainder

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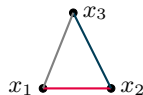
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## Possible Generalizations

- ▶ Gauge theories [Fröb 2016]
- ▶ Curved manifolds
- ▶ Minkowski space?
- ▶ Non-perturbative constructions?
- ▶ Light cone expansion?